



ISO 9001:2015 Certified
Level I Institutionally Accredited

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

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BSIT – 4A

PART 1 – COMPANY NAME

1.1 Company Name

APEX Solutions

1.2 Nature of Business

Software Development and IT Consulting

1.3 Company Vision

To become a trusted software development company that delivers innovative, secure, and scalable digital solutions for businesses across the Philippines and Southeast Asia.

1.4 Company Mission

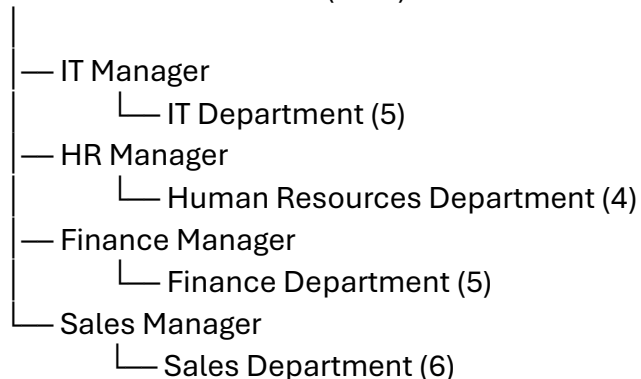
- Develop high-quality software products that solve real business problems.
- Maintain secure and reliable IT systems that support business operations.
- Foster collaboration, innovation, and continuous learning among employees.
- Provide excellent customer service through technology-driven solutions.

1.5 Office Location

4th Floor, Innovatech Building, Central Business Park, Quezon City, Philippines

1.6 Organizational Structure

Chief Executive Officer (CEO)



1.7 Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2 – Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	14	HR, Finance, Sales	Daily office productivity
HW-002	Laptop	6	IT Department, Management	Mobility and remote work
HW-003	Rack Server	1	IT	File sharing, authentication, virtualization
HW-004	Enterprise Router	1	IT	Internet Routing
HW-005	Managed 48-Port Switch	1	IT	Network Connectivity
HW-006	Network Printer	2	Shared	Document Printing
HW-007	UPS 3000VA	2	IT	Power Protection
HW-008	Wireless Access Point	3	Shared	Office Wi-Fi Coverage
HW-009	NAS Storage (16TB)	1	IT	Centralized Storage
HW-010	External Backup Drive (8GB)	2	IT	Offline Backup
HW-011	24-inch LED Monitor	20	All Departments	Employee workstation display

Hardware Justification

Desktop Computers (14)

Desktop computers are cost-effective for employees who work primarily from the office, particularly in HR, Finance, and Sales.

Laptops (6)

The IT team requires portable devices for troubleshooting, server maintenance, and meetings. Management also benefits from mobility.

Rack Server (1)

A single virtualization-capable server can host:

- Active Directory
- File Server
- Backup Services
- Development testing virtual machines

Managed Switch

A 48-port managed switch provides room for future expansion, printers, access points, and additional devices without immediate replacement.

Wireless Access Points

Three access points ensure complete wireless coverage across the office floor and provide redundancy if one device fails.

PART 3 – Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	24H2	Commercial	Employee workstation operating system
Ubuntu Server	24.04 LTS	Open Source	Server operating system
Microsoft Office 365	Latest	Subscription	Productivity applications
Visual Studio Code	Latest	Free	Software development
Git	Latest	Open Source	Version control
GitHub Desktop	Latest	Free	GUI Git management
Oracle VirtualBox	Latest	Free	Virtual machine testing
Google Chrome	Latest	Free	Web browsing and SaaS access
Microsoft Defender	Latest	Included	Endpoint protection
AnyDesk	Latest	Commercial	Remote support
7-Zip	Latest	Open Source	File compression and extraction

Software Justification

Windows 11 Pro

Provides business features such as BitLocker encryption, Remote Desktop, Group Policy, and domain support.

Ubuntu Server

Chosen for its stability, security, long-term support, and low licensing cost.

Development Tools

VS Code, Git, and GitHub Desktop are essential for the software development workflow and collaboration among developers.

AnyDesk

Allows the IT department to provide remote technical support without physically visiting every workstation.

PART 4 – Enterprise Network Inventory

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Fiber Modem	1	Internet connection termination
NET-002	Enterprise Router	1	Routing and NAT
NET-003	Next-Generation Firewall	1	Network security
NET-004	48-Port Managed Switch	1	LAN connectivity
NET-005	Wireless Access Point	3	Wireless network access
NET-006	24-Port Patch Panel	2	Structured cabling management
NET-007	CAT6 UTP Cable (305m box)	2	Network cabling
NET-008	RJ45 Connectors	100	Cable termination
NET-009	Network Rack Cabinet	1	Equipment organization

Network Design Notes

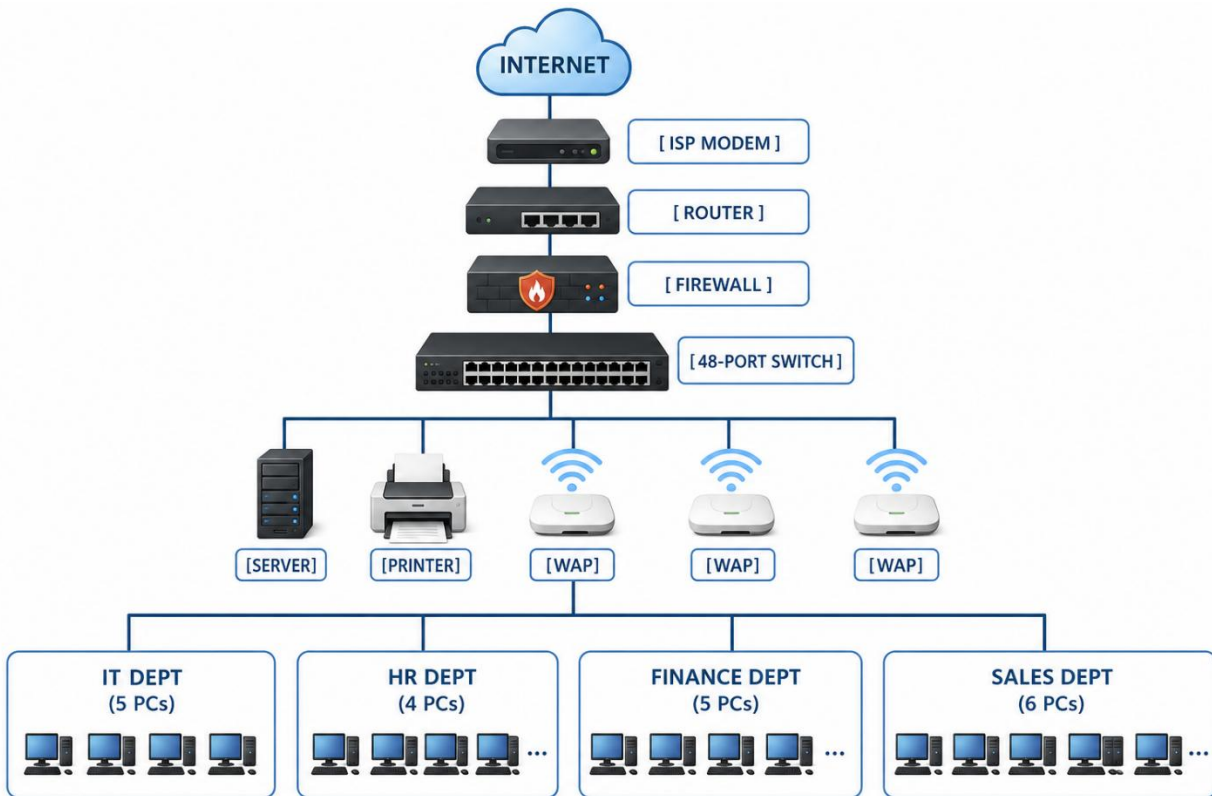
- CAT6 supports 1 Gbps connectivity and future 10 Gbps upgrades over shorter distances.
- The patch panel improves cable organization and simplifies troubleshooting.
- A rack cabinet protects network equipment and improves airflow and maintenance.

PART 5 – Enterprise Network Diagram

Network Topology

Diagram Title: APEX Solutions – Enterprise Topology

Prepared by: Jeffrey Almojela



PART 6 – System Administration Roles

1. Helpdesk Technician

Responsibilities

- Respond to user support requests
- Install and configure workstations
- Troubleshoot hardware and software issues
- Reset passwords and manage user accounts
- Escalate complex issues to senior administrators

Required Skills

- Windows troubleshooting
- Customer service
- Basic networking
- Documentation

- Communication skills

Common Tools

- AnyDesk
- Windows Settings
- Device Manager
- Ticketing systems (Jira, Freshdesk, ServiceNow)

Recommended Certifications

- CompTIA A+
- Google IT Support Professional Certificate
- Microsoft 365 Fundamentals

2. Network Administrator

Responsibilities

- Configure routers, switches, and firewalls
- Monitor network performance
- Implement VLANs and IP addressing
- Maintain internet connectivity
- Troubleshoot network outages

Required Skills

- TCP/IP
- Subnetting
- Routing and switching
- Firewall configuration
- Network monitoring

Common Tools

- Cisco Packet Tracer
- Winbox / MikroTik

- Wireshark
- PRTG Network Monitor
- SolarWinds

Recommended Certifications

- CompTIA Network+
- Cisco CCNA
- MikroTik MTCNA

3. Linux System Administrator

Responsibilities

- Install and maintain Linux servers
- Manage users and permissions
- Configure web, file, and database services
- Monitor server performance
- Automate tasks using shell scripts

Required Skills

- Linux command line
- Bash scripting
- Systemd services
- SSH administration
- Backup and recovery

Common Tools

- Ubuntu Server
- SSH
- Nano / Vim
- Rsync
- Cron

Recommended Certifications

- CompTIA Linux+
- LPIC-1
- Red Hat Certified System Administrator (RHCSA)

4. Cloud Administrator

Responsibilities

- Manage cloud infrastructure
- Deploy virtual machines and storage
- Configure cloud networking and security
- Monitor cloud costs and performance
- Implement disaster recovery solutions

Required Skills

- Virtualization
- Cloud networking
- Identity and access management
- Automation
- Security best practices

Common Tools

- Microsoft Azure Portal
- AWS Management Console
- Terraform
- Docker
- Kubernetes

Recommended Certifications

- AWS Certified Cloud Practitioner
- Microsoft Azure Administrator (AZ-104)

- Google Associate Cloud Engineer

How These Professionals Work Together

In a modern organization, these four IT professionals collaborate closely to maintain reliable business operations. The **Helpdesk Technician** serves as the first point of contact for employees and resolves common technical problems. Issues involving connectivity are escalated to the **Network Administrator**, who ensures that routers, switches, wireless access points, and firewalls operate correctly. Server-related problems are handled by the **Linux System Administrator**, who maintains authentication services, file sharing, backups, and virtualization platforms. The **Cloud Administrator** extends the organization's infrastructure to cloud services for backup, disaster recovery, remote collaboration, and scalable application hosting. Together, these roles create a layered support structure that improves system availability, security, performance, and business continuity.

PART 7 – Infrastructure Recommendations

Internet Provider Recommendation

Primary Connection

- **Fiber Optic Business Plan**
- Minimum **300 Mbps symmetrical**
- Static public IP address
- 99.5% uptime SLA

Reason

A software development company relies heavily on:

- GitHub repositories
- Cloud services
- Video conferencing
- Remote collaboration
- Large file transfers

Symmetrical bandwidth ensures fast uploads for source code and backups.

Recommended Server Specifications

Component Specification

CPU	AMD EPYC / Intel Xeon (8 cores minimum)
Memory	32 GB ECC RAM
Storage	2 × 1 TB NVMe SSD (RAID 1)
Network	Dual Gigabit Ethernet
Power	Redundant Power Supply

Reason

These specifications support virtualization, file services, backups, and future business growth while maintaining reliability.

Backup Strategy

3-2-1 Backup Rule

- **3 copies** of data
- **2 different storage media**
- **1 offsite copy**

Implementation

Backup Type	Schedule	Destination
Incremental	Daily	NAS
Full Backup	Weekly	External Drive
Offsite Backup	Monthly	Cloud Storage

Reason

This approach protects against hardware failure, accidental deletion, ransomware, and natural disasters.

Security Recommendations

Network Security

- Deploy a **Next-Generation Firewall**
- Create separate **VLANs** for each department
- Disable unused switch ports
- Use WPA3 for wireless networks

Endpoint Security

- Enable **Microsoft Defender**
- Apply automatic Windows updates
- Restrict local administrator privileges
- Enable BitLocker encryption on laptops

Physical Security

- Lock the server room
- Restrict access to network racks
- Install CCTV monitoring

Antivirus Recommendation

Microsoft Defender for Business

Advantages

- Integrated with Windows 11 Pro
- Centralized management
- Real-time protection
- Ransomware detection
- Lower operational cost than third-party solutions

Password Policy

Policy	Requirement
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Minimum Length 12 characters

Complexity Uppercase, lowercase, number, symbol

Expiration Every 90 days

Reuse Restriction Last 5 passwords

MFA Required for administrators

Example

APEX@2026!

Strong passwords combined with **multi-factor authentication** significantly reduce the risk of unauthorized access.

Expansion Plan

Year 1 (Current)

- 20 employees
- Single server
- 1 Gbps LAN

Year 2

- Upgrade to **40 employees**
- Add a second virtualization host
- Implement **10 Gbps uplinks** between servers and switches

Year 3

- Introduce **cloud-based disaster recovery**
- Deploy centralized monitoring and logging
- Add redundant internet connections

Reason

Planning for expansion now prevents costly infrastructure redesigns later.

PART 8 – Personal Reflection

Working on this **Enterprise Infrastructure Plan** taught me that system administration involves much more than simply installing computers and connecting cables. I learned how important it is to analyze business requirements before making technical decisions. By studying the needs of each department, I understood that different teams require different hardware, software, security controls, and network access. This project also improved my ability to create professional technical documentation using structured tables, diagrams, and clear explanations.

The most challenging task was designing the **enterprise network topology**. I had to think carefully about how the internet connection, router, firewall, switch, server, printers, wireless access points, and department workstations should be connected logically and securely. Creating a network that is both functional and scalable required critical thinking and attention to detail. I also realized that proper labeling and documentation are essential because future administrators must be able to understand the infrastructure easily.

This project demonstrated why **planning is critical before deployment**. Purchasing equipment without a clear infrastructure plan can lead to unnecessary expenses, compatibility problems, security vulnerabilities, and poor network performance. A well-prepared plan ensures that the organization invests in the correct technologies, allocates resources efficiently, and prepares for future growth. Proper planning also reduces downtime and simplifies troubleshooting because every component has a defined purpose and documented configuration.

Completing this activity has helped me develop skills that are essential for becoming a better **System Administrator**. I practiced infrastructure analysis, inventory management, network design, security planning, backup strategy development, and professional documentation. These are real-world tasks performed by IT professionals in organizations of all sizes. The project also strengthened my confidence in presenting technical information in a clear and organized manner for both technical and non-technical audiences. Overall, this experience gave me a deeper appreciation for the responsibilities of system administrators and prepared me for more advanced infrastructure and networking projects in the future.